

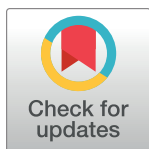
RESEARCH ARTICLE

Test-retest reliability and concurrent validity of knee extensor strength measured by a novel device incorporated into a weight stack machine vs. handheld and isokinetic dynamometry

Pradeep K. Sahu¹, Noel Goodstadt², Arun Ramakrishnan³, Sheri P. Silfies^{1*}

1 Department of Exercise Science, Arnold School of Public Health, University of South Carolina, Columbia, South Carolina, United States of America, **2** Department of Physical Therapy and Rehabilitation Sciences, Drexel University, Philadelphia, Pennsylvania, United States of America, **3** College of Nursing and Health Professions, Drexel University, Philadelphia, Pennsylvania, United States of America

* silfies@mailbox.sc.edu



OPEN ACCESS

Citation: Sahu PK, Goodstadt N, Ramakrishnan A, Silfies SP (2024) Test-retest reliability and concurrent validity of knee extensor strength measured by a novel device incorporated into a weight stack machine vs. handheld and isokinetic dynamometry. PLoS ONE 19(5): e0301872. <https://doi.org/10.1371/journal.pone.0301872>

Editor: Žiga Kozinc, Faculty of Health Sciences, University of Primorska, SLOVENIA

Received: December 29, 2023

Accepted: March 22, 2024

Published: May 22, 2024

Copyright: © 2024 Sahu et al. This is an open access article distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Data Availability Statement: The dataset used and/or analyzed during the current study and description and development of the device are available on the Center for Open Science <https://doi.org/10.17605/OSF.IO/E4GWF>.

Funding: This publication was made possible in part by the Drexel University ExCITE Seed Funding and a grant through the Coulter-Drexel Translational Research Partnership Program (002537). The funders had no role in study design,

Abstract

Background

The current clinical gold standard for assessing isometric quadriceps muscle strength is an isokinetic dynamometer (IKD). However, in clinics without an IKD, clinicians default to using handheld dynamometers (HHD), which are less reliable and accurate than the IKD, particularly for large muscle groups. A novel device (ND) was developed that locks the weight stack of weight machines, and measures forces applied to the machine, turning this equipment into an isometric dynamometer. The objectives of this study were to characterize the test-retest reliability of the ND, determine the within-day and between-days inter-rater reliability and concurrent validity compared with that of the HHD, in healthy volunteers (HV) and individuals with knee osteoarthritis (OA) for measuring knee extensors isometric muscle force.

Materials and methods

29 healthy (age = 28.4 ± 7.4 years) and 15 knee OA (age = 37.6 ± 13.4 years) participants completed three maximum force isometric strength testing trials on dominant side knee extensor muscles on three devices (ND, HHD, and IKD) in two separate sessions by two raters. The maximum force (Fmax) produced, and the force-time series were recorded. Reliability and validity were assessed using Intraclass Correlation Coefficient (ICC), Bland-Altman Plots, Pearson's r , and cross-correlations.

Results

The ND demonstrated excellent test-retest reliability ($ICC_{2,3} = 0.97$). The within-day ($ICC_{2,3} = 0.88$) and between-day inter-rater reliability ($ICC_{2,3} = 0.87$) was good for HHD. The ND showed excellent within-day ($ICC_{2,3} = 0.93$) and good between-day ($ICC_{2,3} = 0.89$) inter-

data collection and analysis, decision to publish, or preparation of the manuscript.

Competing interests: We have read the journal's policy and the authors (ND, AR, SPS) of this manuscript have the following competing interests: A United States patent through Drexel University for the novel device used in this study (Portable Load Testing Device, Patent. No. US 11,099,089). This does not alter our adherence to PLOS ONE policies on sharing data and materials.

Abbreviations: CI, Confidence Interval; Fmax, Mean maximum force; GUI, Graphical user interface; HHD, Hand Held Dynamometer; HV, Healthy volunteers; ICC, Interclass Correlation Coefficient; IKD, Isokinetic Dynamometer; LoA, Limits of Agreement; MDD, Minimal detectable difference; MDabs, Mean Absolute Difference; N, Newtons; ND, Novel Device; NPRS, Numerical Pain Rating Scale; OA, Osteoarthritis; SD_{pooled}, average standard deviation; SEM, Standard Error of Mean; THR, Total Hip Arthroplasty; TKR, Total Knee Arthroplasty;; US, United States.

rater reliability. The Bland-Altman analysis revealed HHD systematic bias and underestimation of force particularly with quadriceps force values exceeding 450 N. Mean differences were found in maximum force between HHD vs. IKD ($MD_{abs} = 58$ N, $p < .001$) but not the HHD vs. ND ($MD_{abs} = 24$ N, $p = .267$) or ND vs. IKD ($MD_{abs} = 34$ N, $p = .051$). The concurrent validity of Fmax ($r = 0.81$) and force-time curve correlation (0.96 ± 0.05) were the highest between the ND and IKD.

Conclusions

The ND's test-retest reliability and concurrent validity make it a potential strength assessment tool with utility in physical therapy and fitness settings for large muscle groups such as the knee extensors.

Introduction

Knee joint injuries account for 39.8% of all sport and 60% of high-school sport-related surgeries in the United States (US) [1, 2]. Quadriceps strength is critical in preventing knee injuries and degenerative joint disease by attenuating ground reaction force [3, 4]. Quadriceps strength impairment is common in osteoarthritis (OA) and post-knee surgery, with deficits that can persist even for years, impacting the ability to perform lower extremity function and sports activities [4–6]. Additionally, longitudinal data indicates quadriceps strength is a robust predictor of functional decline and mortality due to coronary artery disease and chronic obstructive pulmonary disease [7, 8]. Studies showed a positive association of quadriceps strength with dynamic balance, gait speed, stride length, fall risk, general health, and function [9–13]. Evaluation of peak muscle force (strength) is an essential clinical component of assessing muscle function [14]. Moreover, quadriceps strength assessment helps quantify strength deficits, intervention program effectiveness, progression through rehabilitation evaluation for return to sport, and development of normative data [12, 13, 15–17].

Objective strength measurement tools used in clinical practice vary from hand-held dynamometers (HHD) to computerized isokinetic dynamometry (IKD). Utilizing any device on a large scale for clinical practice depends on several factors, such as reliability, validity, affordability, safety, and portability [18]. Reliability evaluates the consistency of repeated measurements, and test-retest reliability specifically addresses consistency across measurement sessions using either the same (*intrarater*) or different (*interrater*) clinical testers [19]. Validity is defined as how well the device measures what it intends to measure. Concurrent validity assesses how well a new device measurement matches an existing criterion.

In clinical practice, muscle strength is often assessed with a HHD [20]. When evaluating knee extensor strength, the examiner typically stabilizes the HHD against the distal anterior tibia, (manually or assisted by a belt) to resist the participant's active knee extension and record the highest amount of force produced during the test [21]. HHDs are portable, inexpensive (~\$1000), and user-friendly, which supports their use in physical therapy, sports, and fitness industries. When larger muscle groups are assessed, HHD application requires changes in practice to account for greater stabilization requirements (for muscles producing more than 15–30 kg force), and alteration in examiner or client position to reduce inconsistency and inaccuracy [21, 22]. Variability in measurement practice is especially problematic for large muscle groups such as knee and hip extensors, as forces produced can reach or exceed 500

newtons [23]. These larger forces rely on the tester's ability to manually resist a patient-generated force; thus, during the use of a HHD, the tester's ability can affect both the reliability and validity of an isometric force reading [21, 24, 25]. Studies have reported differences in maximum force measures of knee muscle extensors and flexors between male and female testers, with the study by Kelln et al. reporting lower reliability with female testers compared to males [26, 27]. To eliminate the issues related to HHD use, the current gold standard for assessing large muscle group isometric strength is the IKD. IKDs are large pieces of equipment that control body segment stabilization, positional consistency, and resistance without the need for the assessor to provide them. Research documenting the reliability and validity of these devices makes them the gold standard for strength testing [28, 29]. Additionally, IKD can measure various variables, including torque, power, and velocity, and the capacity to maintain patient data over an extended period [30]. While a number of companies produce these devices (e.g., Biodex, Computer Sports Medicine Inc., BTE Technologies), they are expensive (>\$40,000) and the associated space requirements restrict the availability of IKDs and often limit their use to research and elite sports training facilities [24, 31]. The IKD can be cumbersome to set up for testing, and training is needed for the efficient operation of the device, further challenging its application.

Reported *intrarater* reliability is moderate to good using conventional HHD protocols for assessing knee extensor muscle strength (examiner provided resistance against a client-generated force). However, *interrater* and test-retest reliability, an indicator of a device's practical use in a clinical setting, is unclear, and significant variations are detected in strength measurement between testers and sessions [27, 32, 33]. Prior work has modified HHD protocols to reduce the influence of the tester's ability to generate an adequate counterforce. Approaches include the use of belts looped around the back leg of the chair to anchor the HHD [34]. Other studies focused on controlling the influence of the tester within the protocol to reduce the error created by poor stabilization of the person or position (providing support by belt and chair frame) [35–38] and leveraging resistance (dynamometer mounted either on a PVC pipe or the back of the chair with a belt) [39, 40]. However, these controlled studies rarely address *interrater* or test-retest reliability. Without between-day test-retest reliability, it is impossible to objectively determine if a client is truly improving their strength in response to an intervention.

Although the modified HHD and force transducer systems have several advantages related to the costs, the major drawbacks in those studies were poor fixation of the HHD pad on the tibial segment [35–44] and lack of proximal stabilization [36–40, 42, 45–47]. Further, many HHDs do not provide feedback during testing [35, 37–41, 43–46], are time-consuming for setting up participants [35, 36, 42], and are not completely tester hands-free measurements [35, 38–41, 43]. A few modified HHD systems are not portable [35, 36, 41, 42, 46, 47], and measuring the strength of other large muscle groups is not provided by the equipment [35, 41, 45–50]. During strength testing, the contact surface of the pad on the tibia may produce pain or discomfort, as reported in a paper [34]. Therefore, it may produce lower force, especially in ongoing painful individuals.

HHD validity studies compared to IKD report correlation values from .43 to .99 across arm and leg muscles, with inconsistent evidence for validation for larger muscle groups [51, 52]. Experiments conducted using modified HHD protocols (external stabilization of the device or person) also provide conflicting results, with some studies indicating that HHD provides a lower force value than IKD, while another study reports overestimation [37, 43]. Despite varying protocol modifications, no standardized protocol has been established that allows clinicians to implement an optimal test efficiently for larger muscle groups. The discussion of limitations of the use of HHD noted in prior studies, particularly when measuring large muscle

group strength, suggests the need for adaptations to testing that allow for more consistent setup, stabilization, and measurement or the necessity for the expensive IKD.

To this end, a novel isometric strength testing device (ND; Patent US 11,099,089) was constructed to address the limitations of the HHD and the current gold standard (e.g., cost, portability, size) while providing reliable and valid strength measures of larger muscle groups. The ND locks down any weight stack, converting it into an isometric system and allowing measurement of the forces applied by the individual through the weight stack cable. Weight stack machines provide improved customization, standardization, and stabilization for individuals during strength testing and eliminate tester manual resistance, and stabilization errors (see the [Materials and methods](#) section for details).

Therefore, the first objective of this study was to determine the within-day test-retest reliability of the ND to assess knee extensors' isometric muscle strength in healthy volunteers (HV) and individuals with knee osteoarthritis (OA). The second objective was to find the within-day and between-day *interrater* reliability of the ND, and to compare it with that of the HHD (with rater manual resistance and participant stabilization). The third objective was to determine the concurrent validity of the ND against the gold standard (IKD) and differentiate that from the concurrent validity of the HHD.

Materials and methods

Participants

This was a prospective observational study conducted at Drexel University. The detailed protocol was approved by the University Institutional Review Board for human subjects (IRB 1610004896), and the participants provided their written consent after discussing potential risks. The sample size was calculated based on the formula reported by Walter et al. for reliability studies (Online calculator-<https://wnarifin.github.io/ssc/ssicc.html>) [53, 54]. To demonstrate the test-retest reliability of ND with expected reliability ($ICC = 0.85$) and a confidence interval ($CI = \pm .1$), 80% statistical power, and two measurements, we calculated the minimum sample size required as 31. We estimated the test-retest reliability of ND to be at least 0.85 as published studies using an HHD demonstrated the *intrarater*, test-retest ICC of knee extensors is 0.78. We believed the ND would surpass this ICC value [21]. We adjusted the sample size needed for this study to 33 participants to account for a 10% dropout rate.

Flyers and posters in the local community and at the university were used to recruit healthy participants (no knee injury or pain) and those diagnosed with knee OA. Participants were recruited between January 3 and October 6, 2017. An online survey, email, and telephone interviews were conducted by the investigators to confirm the participant's eligibility. The inclusion criteria for participants were adults between 18 to 60 years old. The healthy participants had to report no lower extremity injury and no diagnosis of knee or hip OA. Participants with knee OA had a diagnosis of mild to moderate knee osteoarthritis in their dominant leg, were able to walk and climb stairs without an assistive device and had no activity restriction by a physician (cleared for exercise). Potential participants in both groups were excluded if any current or previous spine or lower extremity injury, that significantly affected their daily activities, or recent (last three months) knee or hip surgery on the dominant lower extremity. Other exclusion criteria were neurological deficits, altered sensation, diagnosed depression or anxiety, fatigue disorders, or chronic widespread pain that could affect force production and fatigue of the muscle.

Novel device

The ND consists of structural and electronic elements ([Fig 1](#)). Structural elements are composed of sliding telescopic struts that form the vertical post on which we fixed a 6" right angle

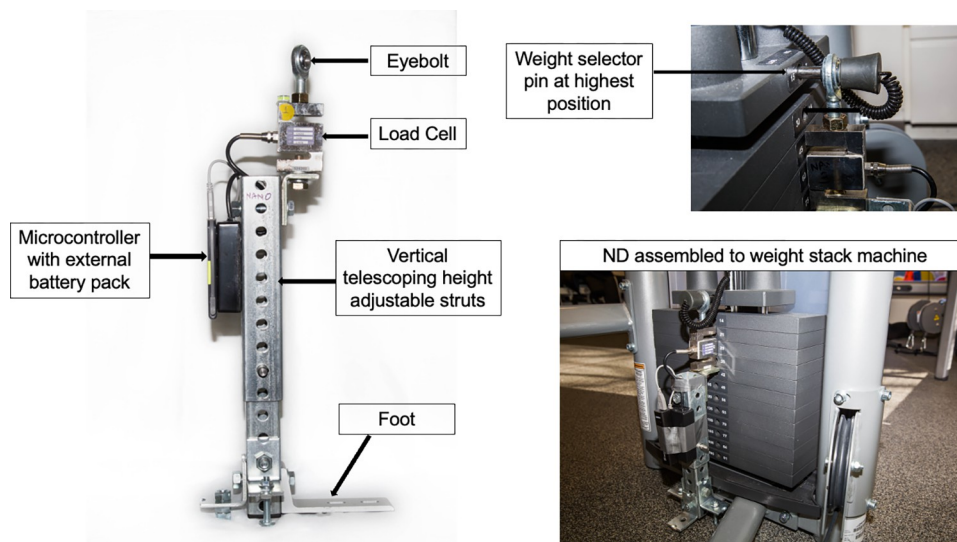


Fig 1. Components of the novel device. The left figure shows the configuration of different parts of the novel device, as described above. The device can be anchored under the bottom of the steel frame of any commercially available weight machine (right bottom figure). The top of the strut has an eyebolt through which the weight selector pin passes and is set at the highest weight stack (right top figure). This locks down the weight stack, converting it into an isometric system and allowing measurement of the forces applied by the individual to lift the weight stack. The load cell registers the application of a tension force to the steel pin. The force is sensed by the digital-to-analog microcontroller attached opposite the load cell. Real-time force data, including the force-time graph from the microcontroller, can be wirelessly displayed and recorded by associated custom software.

<https://doi.org/10.1371/journal.pone.0301872.g001>

foot on the bottom that slides under the weight stack machine's frame. A 1" right angle bracket is secured on the top telescoping strut in which we mount the S-beam load cell (AmCells STL-500lb, tacunasystems.com). An eyebolt was fitted on the top of the load cell, which served as the guide for the pin that locked the weight stack. The electronic components consist of a load cell amplifier (HX711 with 5V excitation) connected to a microcontroller (ATmega328P Nano Board, ARDUINO). Custom firmware was programmed into the microcontroller, allowing it to read the sensor output, convert it to newtons, and transmit it via Bluetooth (HC06) RS232 serial port protocol to a laptop running our custom software. To power these electronics, our initial design used an external 5V rechargeable battery pack, which was later swapped with a 7.4V internal Li-Poly battery and charger pack. A custom graphical user interface (GUI) program running on the laptop/tablet was written in LabVIEW (version 8.6, National Instruments, Austin, TX) to connect, display, and save real-time data. A PC application was developed that added functionality, including subject profile, report generation, and several graphing improvements. For additional information regarding the development and design of the device, see <https://doi.org/10.17605/OSF.IO/E4GWF>.

Procedures

Fig 2 provides an overview of the measurement protocol, which was conducted in two separate sessions 3–14 days apart. The total time needed to complete each session was approximately 1.5 hours. Session one included: assessment of demographics and anthropometric data [age, sex, height (cm), weight (kg), thigh and lower leg segment lengths (cm), and leg dominance (the leg used to kick a soccer ball with maximal effort)]. Participants completed the Saltin-Grimby Physical Activity Level Scale (SGPALS) questionnaire [55]. This is a 4-level scale that measures leisure time physical activity, with the lowest score 'of 1' representing "physically

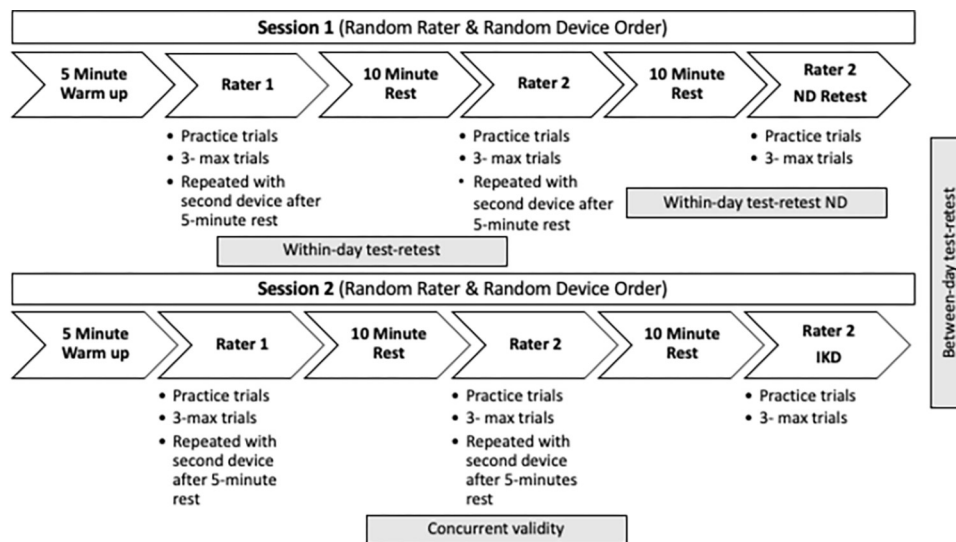


Fig 2. Measurement protocol. This is a schematic of the two-session testing protocol, with an indication of the datasets used for each analysis.

<https://doi.org/10.1371/journal.pone.0301872.g002>

inactive” and the highest score ‘of 4’ representing “regular hard physical training for competitive sports.” This tool previously showed a positive association with cardiovascular risk factors, mental health, and predictors of cerebrovascular diseases [56–58]. This self-reported scale has demonstrated good reliability and validity in a large-scale adult population [59]. The participants with knee OA were also asked to note their side of pain and the intensity of pain recorded using an 11-point numeric pain rating scale (NPRS) [‘0’ corresponding to “no pain” to ‘10’ represents “maximum intensity of pain”]. This scale has demonstrated acceptable ranges of reliability [60]. During each session, raters collected dominant leg knee extensor muscle strength using the HHD and ND with device testing and rater order randomized *a priori*.

In session one, the second rater completed a second measurement series using the ND to obtain the ND’s test-retest reliability. The strength measurement using the IKD was always performed in the second session and by the second rater after the measurements using the ND and HHD were completed. Two licensed physical therapists and one exercise scientist (three males) were trained as testers on all three devices. All testers were familiar with and used HHD and IKD testing devices.

Pre strength-testing set-up

The participants were asked to wear shorts. They were positioned in the knee extension weight stack machine, and the axis of the device was aligned with the participant’s knee axis (dominant leg). The shin pad was kept as low on the tibia as possible, proximal to the lateral malleolus, with the participant still able to dorsiflex their ankle to a neutral position. Anthropometric measurements were taken in this position: thigh length (greater trochanter to lateral femoral condyle), leg length (knee joint line to the marked point for tibial pad), tibial pad location (mid-point on lateral malleolus to the tibia pad mark) and used to standardize the location of resistance on the distal tibia across the devices.

The participants performed 5 minutes of cycling on a stationary bike (75–100 watts), with no resistance, for warm-up before conducting measurements. As part of the warm-up, the participants performed one sub-maximal trial (50–75% of possible maximum effort) and 1–2 trial

(s) with maximum intensity on each device. During the warm-up and practice trials, participants were instructed to build their force gradually and then hold the maximum effort. The instruction was “once the tester says start, build the force to reach the maximum effort in 1–2 seconds and hold the maximum until the tester says stop” (which was standardized to 5 seconds). No visual or verbal feedback was provided during warm-up or testing, and a 5-second contraction time was used to obtain a maximum force value [61]. A one-minute rest was provided between the sub-maximal and maximum trials. Then, three maximum isometric force ($F_{\max}$) trials were performed with a 1-minute rest between the trials to avoid fatigue [62]. A 5-minute rest period was given between the devices and 10 minutes between the raters (Fig 2).

Testing protocol using the HHD

After the participants and the HHD were set up for testing (Fig 3), the participants were instructed to keep their trunks straight and hold the sides of the plinth to avoid any backward movement of the trunk. HHD force application by the testers was at a right angle to the tibia. Before starting the measurement, the HHD force was reset to zero. A second investigator recorded the maximum force from the HHD screen, and the time-series data were recorded at 40Hz.

Testing protocol using the ND

Set-up of the ND. Attaching and anchoring the ND to the weight stack requires aligning the device with the center of the weight stack and sliding the adjustable base under the frame of the weight machine. The device should be lifted so the adjustable base of the unit is in contact with the base of the weight machine frame prior to pushing the weight stack pin through the eyebolt of the ND. Gross adjustment to align the eyelet with the hole of the top plate on the weight stack is through a pin and slide mechanism. Fine adjustment is via a nut attached to the eyelet to adjust the height within fractions of an inch for the pretension of the load cell. The entire process takes less than 5 minutes. Further procedures to run the test require the patient being fitted into the machine to align the joint being tested to the axis of the machine lever arm, measuring the distance between the axis of rotation and the point of force development, and setting up the test information required. Test information includes auto-generated patient ID, test type and the number of trials, notation of the involved side, patient’s diagnosis, test limb, and joint angle. Other information that can be saved is turning on enabling of threshold and percentage of threshold and providing notes on set-up or other important information. Once saved, the Bluetooth is connected, and the test is ready to run.

Testing by using ND. Once the participants and the ND were set (Fig 4), the load cell force output was set to zero. The force data were collected by the custom-built software at 10 Hz. The software was set to account for the machine’s CAM design at 90 degrees of knee flexion. The second investigator recorded the maximal force values, and time-series data were recorded.

Testing protocol using the IKD

The participants were asked to sit in the IKD machine chair (Cybex, Humac Norm, US; Fig 5). Force was recorded (kg), and the device was set to isometric mode with maximum force recorded and time-series data collected at 100 Hz. $F_{\max}$ and time-series data were converted to Newtons (N) for analysis.

Statistical analyses

Statistical analysis was conducted using the SPSS version 28 (Armonk, NY: IBM Corporation). Demographics (Mean, SD, N) were calculated for participants’ general characteristics. The

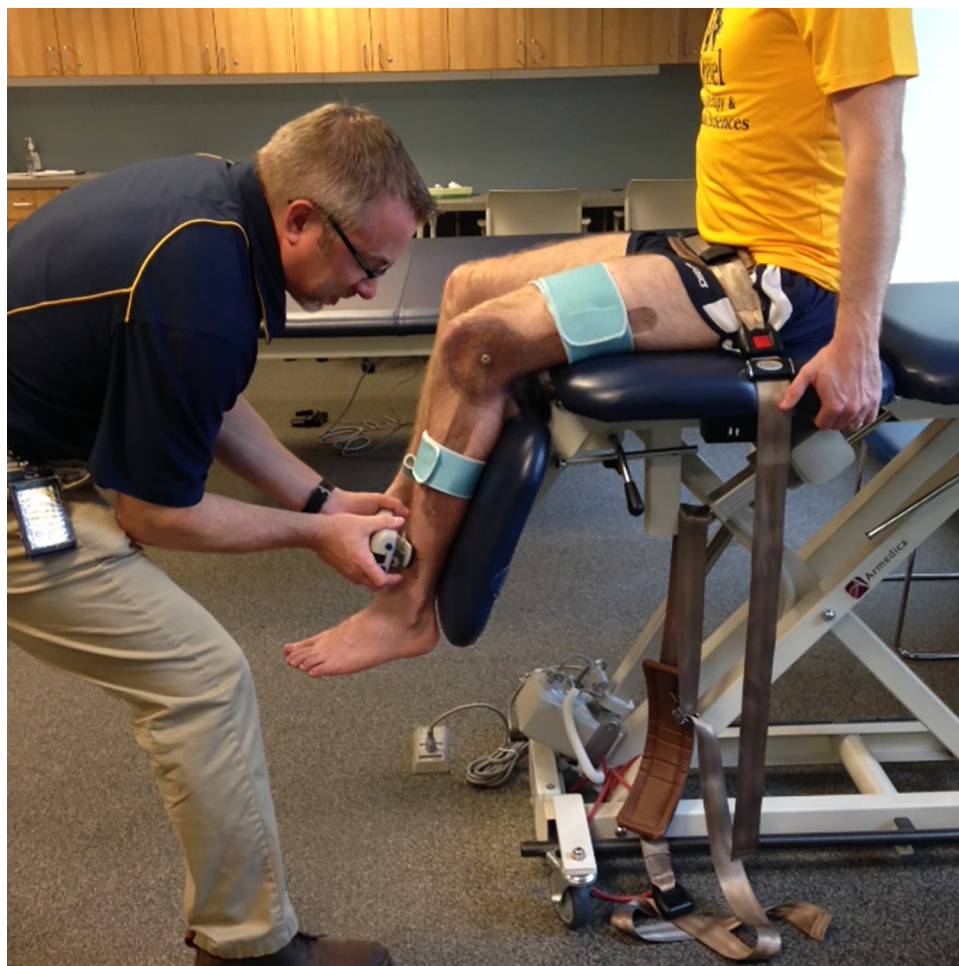


Fig 3. Set up for HHD testing. Photo of the set-up for measuring isometric knee extensor strength on the dominant side using a commercially available HHD (Lafayette Instruments, US, model no. - 01165). The participant was positioned in high sitting at the end of a treatment table with legs hanging off the side at 90° of knee flexion and posterior thigh in complete contact with the treatment table. The proximal thigh was fixed to the treatment table by a mobilization belt. To ensure the angle of the knee joint was at 90° before each trial, a universal goniometer was attached using Velcro with the axis at the lateral femoral condyle, fixed arm directed towards the greater trochanter, and movable arm towards the lateral malleolus. The HHD units were set to Newtons (N) and measurement time to 5 seconds before placing the device on the participant's tibia. The HHD is strapped to the tester's dominant hand so that the tester can place the measurement side of the HHD on the previously marked point of the participant's tibia. The plinth height was raised to allow the tester to stabilize their elbow at 90 degrees of flexion against their body when providing resistance perpendicular to the tibia against the participant's knee extensor force.

<https://doi.org/10.1371/journal.pone.0301872.g003>

normality of data was checked by the Shapiro-Wilk test. The average value (Mean, SD) of three maximum force trials for each test was used for analyses. Data were analyzed as all participants combined and separately by group (HV, OA). Session one data was utilized for within-day *interrater* reliability analyses and ND reliability (*intrarater*, second rater data) using a two-way mixed random absolute agreement model (ICC_{2,3}). Data from both sessions was used for *interrater* between-day reliability of the HHD and ND using a two-way random model absolute agreement (ICC_{2,3}) [63]. ICC values were interpreted as follows: < 0.5; poor reliability, 0.5 to 0.75; moderate reliability, 0.75 to 0.90; good reliability, and > 0.90; excellent reliability [64]. The standard error of the mean assessed absolute reliability (SEM) = $SD_{\text{pooled}} \times \sqrt{1 -$

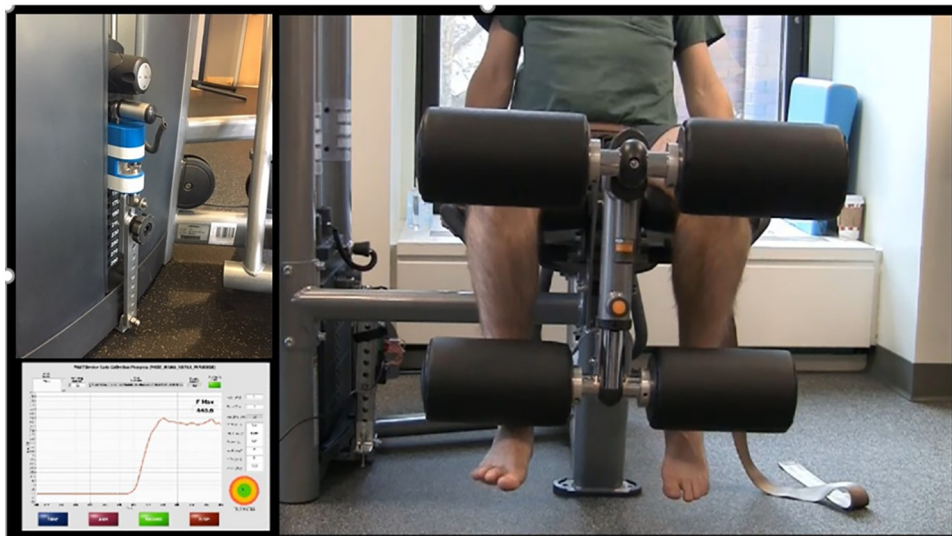


Fig 4. Set up for ND testing. Photo of the set-up testing knee extensor strength with the ND (left top panel) and the accompanying generated force-time curve (left bottom panel; X-axis is time, and Y-axis is the force in newtons). Participants were seated in the knee extension weight stack machine (Life Fitness, Illinois, USA; right panel). The axis of the participant's knee joint (tip of lateral femoral condyle) and the machine axis were aligned. Participants were positioned so that the seat fully supported their thighs, and pillows were used when necessary to ensure a trunk-neutral position. The strapping, belt fixation, and goniometer placement procedures were the same as the HHD protocol. Knee flexion was aligned to 90°, and participants were asked to hold the side handles of the chair similar to the HHD protocol to maintain the neutral trunk alignment. The tibial pad was located at the mark on the tibia. The ND was inserted into the weight stack, leveled, and adjusted so no slack was present in the cable system.

<https://doi.org/10.1371/journal.pone.0301872.g004>

ICC), where SD_{pooled} is the mean of both SDs. Minimal detectable difference (MDD) was calculated using the formula $MDD = SEM \times z \text{ score} \times \sqrt{2}$, and the MDD 95% ($z \text{ score} = 1.96$).

The data from the second testing session were used to analyze concurrent validity. To assess systematic bias, Bland-Altman plots, including the limits of agreement (LoA), were created between each pair of devices for all participants, and HV and OA groups separately in MATLAB (The MathWorks, Inc., Natick, MA, USA) [65]. One-way repeated measure ANOVA was implemented to compare the maximum force value across the three devices, and a pairwise comparison was performed to obtain the differences between each pair of devices. Pearson correlations (r) of maximum force values between each pair of devices were calculated and interpreted as negligible (0.0–0.10), weak (0.10–0.39), moderate (0.40–0.69), strong (0.70–0.89) or very strong correlation (0.90–1.00) [66]. Force-time curves representing force generation patterns between each pair of devices for each participant were down-sampled to 10 Hz, spline interpolated, and correlated in MATLAB. All statistical analyses were two-tailed, and the significance level (p) was < 0.05 .

Results

Demographics and general characteristics

Twenty-nine HV (age: 28 ± 7) and 15 OA (age: 38 ± 13) participants (total $n = 44$) were deemed eligible and participated in this study. For the reliability analysis, six healthy individuals' data were excluded from the study due to device errors, and therefore, 23 HV and 15 OA participants were considered for reliability analysis. All 29 HV and 15 OA participants' data were used for validity testing. A detailed description of the participant demographics, anthropometric data, physical activity levels, and testing data pain intensity are presented in Table 1.



Fig 5. Set up for IKD testing. Photo of the set-up for testing knee extensor strength using the IKD. The participants were seated in the IKD machine chair (Cybex, Humac Norm, US), with the lateral epicondyle of the tested side femur centered with the IKD's input arm axis by adjusting the chair's position. The hip knee adapter stabilizing pad was kept against the participants' tibia at the tibial marking by adjusting the knee/hip adapter and stabilizing pad until the bar was at the correct height. The investigators strapped the torso, thigh, and tibial harness to provide stability during the test. The knee angle (90° of flexion) was manually adjusted using the software and confirmed with a universal goniometer.

<https://doi.org/10.1371/journal.pone.0301872.g005>

Reliability

Test-retest reliability of novel device. The test-retest reliability of the ND (Table 2) $ICC_{(2,3)}$ was 0.97 (with narrow CIs) for all participants and did not change after group separation. The SEM was 53 N, and MDD_{95} was 148 N when all participants were included.

Within- and between-day interrater reliability. Table 3 provides the within-day and between-day *interrater* reliability (ICC, CI), including the group mean Fmax values, SEM, and MDD_{95} of all participants and HV and OA groups separately.

Within-day interrater reliability. The within-day *interrater* reliability of all participants was $ICC_{2,3} = 0.88$ for the HHD and $ICC_{2,3} = 0.93$ for the ND. In all analyses, the HHD had a

Table 1. Demographics and clinical characteristics of all participants and separated by group.

Participants Characteristics	Total (N = 44) (Mean $\pm$ SD/N)	HV (N = 29) (Mean $\pm$ SD/N)	Knee OA (N = 15) (Mean $\pm$ SD/N)
Age (years)	31.6 $\pm$ 11.2	28.4 $\pm$ 7.4	37.6 $\pm$ 13.4
Height (cm)	168.2 $\pm$ 8.6	168.3 $\pm$ 6.9	168 $\pm$ 11.4
Weight (kg)	73.8 $\pm$ 20.6	68.3 $\pm$ 11	84.4 $\pm$ 29.8
Gender (F, M)	28 F, 16M	19 F, 10 M	9 F, 6 M
Dominant leg (Right)	39	26	13
NPRS (0–10 cm)	0.4 $\pm$ 1.2	0 $\pm$ 0	1.1 $\pm$ 1.8
SGPALS	2.9 $\pm$ 0.8	3.1 $\pm$ 0.7	2.5 $\pm$ 1
Days between sessions	10 $\pm$ 3	12 $\pm$ 3	9 $\pm$ 4

OA = Osteoarthritis, NPRS = Numerical pain rating scale, SGPALS = Saltin-Grimby Physical Activity Level Scale

<https://doi.org/10.1371/journal.pone.0301872.t001>

wider ICC CI than the ND. Separate group analysis reveals that the reliability of the HHD was higher in the OA group compared to the HV group. In contrast, reliability of ND was not different between OA and HV groups. The results also showed that the HHD has a higher ICC value in the OA compared to the HV group. When all participants were considered, the measurement from ND had a greater SEM and MDD₉₅ compared to HHD, principally due to the larger pooled between subject SD when using the ND. Similarly, in both the groups, the SEM and MDD₉₅ values were lower for HHD than ND.

Between-day interrater reliability. The *interrater* reliability values were good for both the HHD (ICC_{2,3} = 0.87) and ND (ICC_{2,3} = 0.89), across all participants. However, in the separate group analysis, both devices demonstrated lower reliability in the OA group, than the HV. In all analyses, the HHD had a wider ICC CI than the ND. Similar to the within-day *interrater* reliability, all participants had lower SEM and MDD₉₅ values with the HHD than the ND.

Validity

One-way repeated measure ANOVA (Table 4) indicates a statistically significant difference in maximum force generated across the three devices ($p < 0.001$). Pairwise probability comparison demonstrates differences between the HHD and IKD ($p < 0.001$, MD_{abs} = 58 N), while no statistically significant difference was found between the HHD vs. ND and ND vs. IKD. Pearson correlations reveal the highest correlation between ND vs. IKD ($r = 0.81$) compared to the correlation between HHD vs. ND ($r = 0.72$) and HHD vs. IKD ($r = 0.74$). Force-time curves (representing force generation patterns across time) for each participant (Fig 6) were

Table 2. Test-retest reliability of novel device (within-day, intra-rater) of all participants and separated by group.

Subjects	Fmax (N) Test 1	Fmax (N) Test 2	SD _{pooled} (N)	ICC (95% CI)	SEM N (%)	MDD ₉₅ N (%)
All	414.2	425.1	307.1	.969 (.940, .983)	53 (8.1)	148 (22.3)
HV	464.5	474.3	194.2	.983 (.959, .993)	34 (5.0)	93 (13.8)
OA	350.3	349.7	142.2	.947 (.848, .982)	29 (10.4)	80 (29)

All = All participants, HV = Healthy volunteers, OA = Osteoarthritis, ND = Novel Device, Fmax = Mean maximum force, N = Newtons, % = Percent of Mean Force, SD_{pooled} = Average standard deviation, ICC = Interclass Correlation Coefficient, CI = Confidence interval, SEM = Standard error of mean, MDD₉₅ = Minimal detectable difference

<https://doi.org/10.1371/journal.pone.0301872.t002>

Table 3. Within-day and between-days interrater reliability of the hand-held and novel devices for all participants and separated by group.

Within-day Interrater Reliability							
Subject	Device	Fmax (N) Rater 1	Fmax (N) Rater 2	SD _{pooled} (N)	ICC _{2,3} (95% CI)	SEM N(%)	MDD ₉₅ N(%)
All	HHD	406.6	395.4	95.9	.881 (.631, .882)	33 (8.1)	92 (22.5)
	ND	430.4	411.1	177.2	.928 (.864, .962)	48 (11)	132 (30.6)
HV	HHD	443.1	428.8	75.7	.798 (.531, .914)	34 (7.6)	94 (21.2)
	ND	481.1	464.8	162.1	.905 (.779, .960)	50 (10.3)	139 (28.7)
OA	HHD	372.5	361.8	79.6	.892 (.690, .962)	26 (7)	73 (19.4)
	ND	380.9	356.3	155.5	.935 (.819, .977)	40 (10.3)	110 (28.8)
Between-day Interrater Reliability							
All	HHD	409.2	416.1	86.7	.869 (.775, .924)	32 (7.5)	87 (20.9)
	ND	440.9	438.8	178.6	.885 (.802, .933)	61 (13.8)	168 (38.2)
HV	HHD	427.6	439.5	84.1	.870 (.749, .933)	31 (6.8)	85 (19.1)
	ND	469.4	441.9	178	.928 (.859, .963)	48 (10.8)	133 (30.1)
OA	HHD	369.1	365.0	75.3	.788 (.402, .924)	35 (9.4)	97 (26.3)
	ND	378.9	431.9	172.6	.780 (.416, .919)	81 (18.7)	224 (51.9)

All = All participants, HV = Healthy volunteers, OA = Osteoarthritis, HHD = Handheld dynamometer, ND = Novel device, Fmax = Mean maximum force, N = Newtons, % = Percent of mean force, SD_{pooled} = Average standard deviation, ICC = Interclass Correlation Coefficient, CI = Confidence interval, SEM = Standard error of mean, MDD₉₅ = Minimal detectable difference

<https://doi.org/10.1371/journal.pone.0301872.t003>

correlated between each pair of devices and then averaged. We found the highest correlation average between the ND vs. IKD (0.96 ± 0.05).

Bland-Altman plots (Fig 7) indicate systematic bias of the HHD when compared with IKD and ND. The plots also display proportionate bias when comparing HHD with IKD and ND as the HHD progressively increases the underestimation of participant force starting around 450 N. Systematic bias was slight, and no proportionate bias was observed between ND and

Table 4. Statistical procedures and outcomes for concurrent validity testing all participants.

Statistical Procedure	HHD vs ND	HHD vs IKD	ND vs IKD
One-way repeated measure ANOVA	0.267	<0.001	0.051
Pairwise comparison (p-value)			
Pairwise mean difference (HHD-ND; HHD -IKD; ND-IKD) of Fmax (Newtons)	-24	-58	-34
Pearson correlation (r)	0.717	0.742	0.809
(95% CI)	(0.534, 0.836)	(0.571, 0.851)	(0.674, 0.892)
Force-Time curves Correlation (r) (Mean±SD)	0.82 ± 0.19	0.83 ± 0.19	0.96 ± 0.05

ANOVA = Analysis of covariance, HHD = Hand-held dynamometer, ND = Novel Device, IKD = Isokinetic dynamometer, $p < 0.05$ for statistical significant threshold, r = Pearson correlation coefficient, CI = Confidence interval

<https://doi.org/10.1371/journal.pone.0301872.t004>

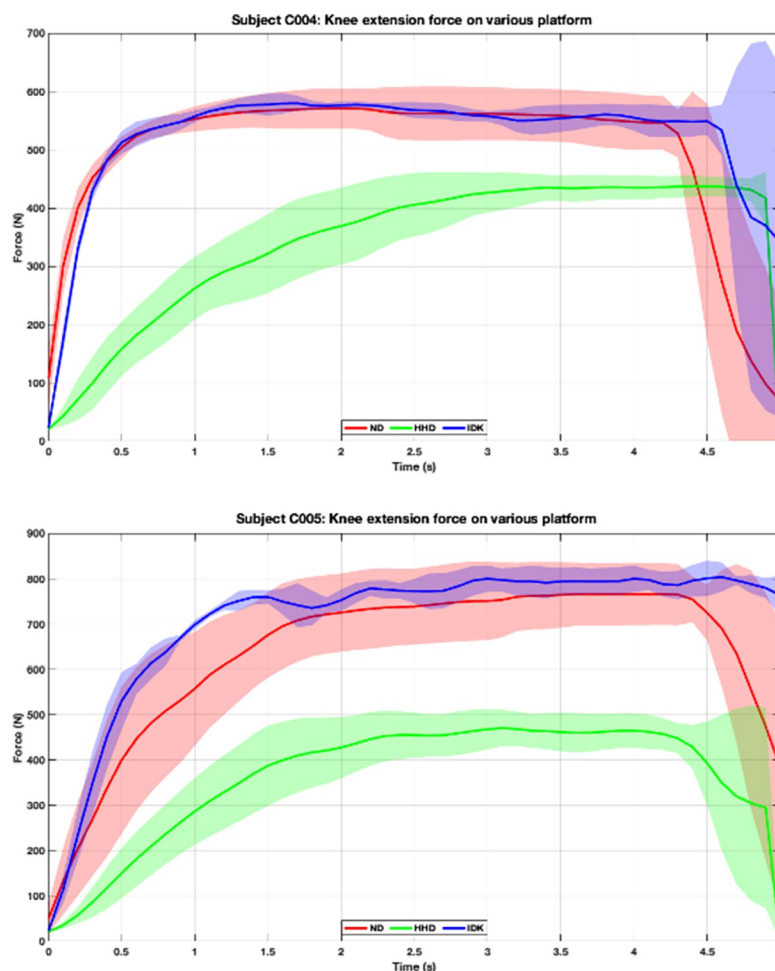


Fig 6. Example force-time curves. Representative time-force curves, averaged across three trials, for two participants (time on the X-axis and force on the Y-axis). The force-time curve mean and standard deviation (shadow) of the ND (red) are similar to those of the IKD (purple). However, the HHD force-time curve (green) differs from those of ND and IKD.

<https://doi.org/10.1371/journal.pone.0301872.g006>

IKD. The LoA ranges were fairly large between all the devices due to the sample's between-subject variability.

Discussion

The overall results denote that the ND has excellent device reliability for measuring knee extensor isometric strength within a session using the same rater. The within-day and between-day *interrater* reliability of the HHD was good, whereas good to excellent reliability was detected for the ND. The findings indicate that the ND demonstrates higher concurrent validity against the gold standard (IKD), while a significant difference in force and proportional bias was observed between HHD and IKD.

Test-retest device reliability

To objectively measure and document the response to certain rehabilitation interventions such as exercises and manual therapy, muscle strength assessment using a HHD or IKD is common. For those clinics without an IKD, testing larger muscle groups presents a challenge

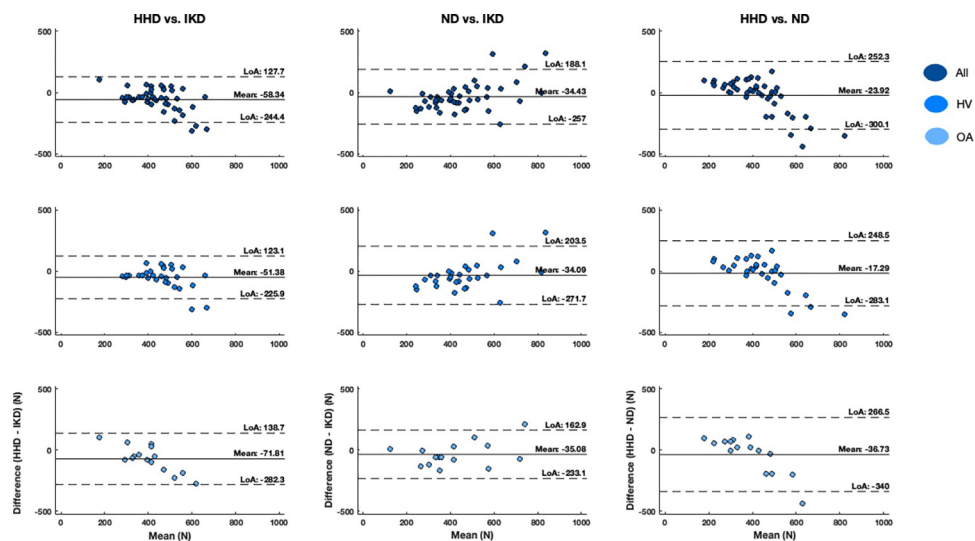


Fig 7. Bland-Altman plots. These represent the mean difference, p-value, limits of agreement (LoA), and bias between each pair of devices. The upper row plots include all participants, the middle row the HVs, and the bottom row knee OA participants. Each graph represents the mean difference at the center (solid line), the upper and lower limits of agreement, (dashed lines), and the dots represent the between device difference in Fmax for each participant.

<https://doi.org/10.1371/journal.pone.0301872.g007>

for obtaining accurate and consistent results. Study findings consistently report that the removal of rater-generated resistance from the strength testing protocol improves within-day and between-day reliability [29, 45–47]. Our ND eliminates the need for tester resistance by utilizing equipment already found in most clinics. We inserted a tension load cell into a weight stack machine and took advantage of the machine's adjustability, ease of patient stabilization, and ability to select joint positioning thus creating a hands-free strength testing unit. Our ND demonstrated excellent *intra-rater* test-retest reliability ($ICC_{2,3} > .90$) across both HV and participants with OA.

Within-day reliability in healthy individuals

Most studies using HHD for knee strength testing report only within-day *intra-rater* reliability. Studies of conventional HHD protocols (manual resistance) reported good to excellent within-session *intra-rater* test-retest reliability ($ICC = 0.70–0.92$) for knee strength in healthy young adults [21, 27]. Researchers have approached the problem by using a belt to anchor the HHD or mounting it to a frame attached to a bench or chair and demonstrated that using their frame-mounted system resulted in good to excellent ($ICC = 0.89–0.98$) within-day *inter-rater* reliability [34, 45–47]. In a within-day test-retest study that mounted the HHD to PVC, they reported excellent *intra-rater* ($ICC = 0.93$) reliability in 15 young, healthy individuals [39]. A study by Sung et al. modified their HHD protocol by anchoring the HHD to a frame but had the tester manually stabilize the pelvis during testing of knee extensor strength and reported excellent *intra-rater* test-retest reliability ($ICC = 0.98$) [41]. The ND in our study demonstrated excellent ($ICC_{2,3} > .90$) within-day *intra-rater* test-retest reliability in both HV and OA participants, and the narrow CIs suggest the ability of the device to capture reliable data even in participants with greater strength.

In clinics, multiple examiners often gather data for intervention progression decisions and documentation. Variability among clinical testers has been commonly reported in the literature. The potential for inconsistency among testers necessitates a device that minimizes

sources of error to make it suitable for use in research and practice [67, 68]. Therefore, a majority of this study specifically focuses on *interrater* reliability to illustrate a worst-case scenario within the clinic. We found that the within-day *interrater* reliability of the HHD was good when the HV and OA participants' data were analyzed together. However, the reliability of the HHD was lower in HV participants compared to knee OA, implying that the amount of force generated by the participants plays a crucial role in HHD reliability. Further, a wider CI was observed in the HV compared to OA individuals, again suggesting the testers' strength could influence measurement *interrater* reliability. We found a study by Sung et al., that reported within-day *interrater* reliability with a HHD as excellent ($ICC = 0.98$) which they achieved by mounting the HHD to an external frame [41]. This suggests it is necessary to move beyond manual resistance to a system that augments participant and joint positioning and dynamometer stabilization.

Between-day reliability in healthy individuals

In clinical settings, knee extensor strength assessment is an essential indicator of response to rehabilitation and requires testing during different treatment sessions. Variability in quantitative measurements increases, and the agreement reduces when testing on separate days. Therefore, addressing the between-day *interrater* reliability contributes to the uniqueness and acceptability of devices. Only a few studies reported between-day reliability of conventional HHD testing and only in healthy individuals with studies reporting good to excellent *interrater* ($ICC = 0.77$ – 0.98) reliability between sessions [27, 32, 33, 69]. In parallel, the present study found good reliability using the HHD with manual resistance in HV, when protocol repeatability, participant stabilization, and tester positioning were optimized. On the other hand, the ND demonstrated excellent between-day *interrater* reliability in HV with a tight CI and good reliability in participants with OA. Comparing our SEM and MDD_{95} values to prior studies reporting between-day reliability is complicated by the tester differences and participant samples used. We found one paper reporting between-day *interrater* reliability for knee extensors using a dynamometer and strap and a few between-day *intrarater* studies reporting these values and only in healthy participants. Morin et al. reported a between-day *interrater* reliability of ($ICC = 0.95$) with an SEM 1.69% (2.47 Nm) and MDC_{95} 4.69% (6.84 Nm), which represent a lower percent of the mean than our reported values [33]. Mentiplay et al., reported between days *intrarater* values ranging between 7.7 to 8.7% of the mean force for their SEM and 21.4 to 24.2% of the mean for the MDC_{95} using a HHD with manual resistance for testing knee extensor strength [32]. These values compare to the SEM of 6.8% and MDD_{95} of 19.1% in our study for *interrater* HHD. In studies where the HHD was fixed by a belt, a lower percentage of mean values were reported for the SEM 5.21% and MDC_{95} 14.78% [38]. Martins et al. also performed a between-day *intrarater* study reporting an SEM of 12% and MDD_{95} of 34% for a HHD mounted to the arm of an IKD system [36]. This situation more closely matches our ND in which we found an SEM of 10.8% and MDD_{95} of 30.1%. The higher MDD_{95} in our study is partially driven by our HV large between-subject variance, which we attribute to the concurrent validity demonstrated by the ND. For comparison, we found one study that reported percent of mean force SEM and MDD_{95} values of 5.7% and 15.7%, respectively, when measuring knee extensor strength with an IKD [32].

Within and between-day reliability in a clinical population

Few studies assessing knee strength in a clinical population have been conducted that reported *intra*- and *interrater* reliability when using a HHD. Studies testing *intrarater* knee extensor strength within-day were located using participants with OA ($ICC = 0.92$ – 0.97) and

hematological malignancy (ICC = 0.94) [40, 70]. Another study reported excellent knee extensor muscle strength reliability using a HHD within-day *intrarater* (ICC = 0.98) and *interrater* reliability (ICC = 0.95–0.96) in an unspecified rehabilitation sample [71]. Additional studies conducted in arthritic populations (e.g., TKR, THR, OA) modified their HHD protocols and reported that their redesigned system enhanced (ICC = 0.92–0.99) *interrater* reliability during a single session [35, 40]. The potential drawback of developing individual methods for improving HHD reliability of strength measurements in larger muscle groups, is the potential lack of consistency across raters and the time needed to set up the testing protocol within the clinic setting. The present study found good within or between-day *interrater* reliability using the HHD in a tightly controlled protocol, while the ND demonstrated excellent within-day or good between-day *interrater* reliability in participants with OA, with easier setup.

Concurrent validity

Most prior research identified that HHDs underestimate the force compared to the gold standard (IKD). A study in older adults found approximately 21 Nm and 27 Nm reduced torque by conventional HHD compared to IKD on the left and right knee extensors, respectively [69]. About a 15 Nm difference in knee extensor torque was found even after altering the participant's position to supine lying with the knee in 35° of flexion, which provided better leverage to the tester, offering resistance [42]. A few studies used belt-fixed HHD to minimize the role of the tester's resistance and discerned that the HHD either underestimated or over-estimated the force compared to IKD [37, 43, 44]. Bohannon et al. reported a 35 Nm underestimation in knee extensor torque in HV [43]. Similarly, another study detected 230 N less knee extensor force obtained with the HHD in healthy individuals [44]. In contrast, Lesnak et al. showed belt stabilized HHD overestimates torque 19.4 Nm on average, and the difference was above the reported MDD [37]. However, the lack of adequate proximal segment stabilization owing to the possibility of synergistic muscle groups contributing to the recorded force production could account for overestimations. In addition, a higher possibility of dynamic movement is suspected during some modified and manually resisted HHD strength protocols, risking measurement consistency and validity. We observed significantly lower forces generated during HHD (58 N) testing compared with IKD, while no statistically significant difference in force was found between the ND and IKD (34 N). The differences in force generated by HHD and ND compared to IKD were below MDD₉₅, and the ND more closely matched that of the gold standard. One of the primary reasons behind the higher SEM and MDD₉₅ for the ND is the greater variability in participant generated force, particularly in the HV, as the ND does not appear to demonstrate a ceiling effect or proportional bias as noted in the HHD vs IKD Bland-Altman plots. The HHD increasingly underestimated the force generated by the participants starting around 450 N. This is a major limitation of the use of a HHD with manual resistance in clinical settings, especially when measuring the strength of large muscle groups and in participants with greater strength [43]. In contrast, the ND did not display proportionate bias, suggesting its validity for application in large muscle group testing and its capture of true isometric strength.

We found a strong correlation among maximum force values obtained by all the instruments; however, the highest correlation coefficient was demonstrated between the ND and IKD. Earlier studies of HV focusing on younger and older participants documented a strong correlation (0.78–0.92) with IKD either by using the conventional HHD technique or when placing the HHD on the tibial pad of a weight stack [32, 36, 42]. The possible reason behind a greater correlation between HHD and IKD in those studies, in contrast to our research, might be attributed to methodological differences such as manual resistance, unstandardized

stabilization of the proximal segment, and or positioning of the participants. Another factor likely contributing to a lower correlation of the maximum force produced in our study is between participant variability in our data, as both HV and the clinical population were included in the validity analysis. A potential reason for the lower force generation and less agreement when using the HHD could be pain or discomfort reported at the tibial contact site with the HHD [34]. In our study, 34/44 individuals reported contact site pain during HHD testing, whereas only 19/44 and 11/44 participants reported tibia contact discomfort during testing with the ND and IKD, respectively.

Diverging from other currently available literature, we also assessed the force-time series correlation. We found the highest correlation between ND and IKD with a small SD which illustrates that the force generation pattern matched well with the IKD. This finding suggests that in those clinics without an IKD, the ND could be an efficient means for evaluating rate-of-force/torque development, which is a variable of interest in strength recovery metrics [72].

In fitness settings and clinics without IKD, but with weight stack machines, use of the ND can expand strength testing capabilities of larger muscle groups without significant increases in practitioner time or client burden. The typical testing time using the ND, inclusive of set up, 2–3 warm-up repetitions, and 3 test trials on each limb, is generally 15–20 minutes and allows for more consistent testing positions and easier stabilization of clients than using a HHD. The reduced ND setup time and less complicated software shorten the testing time by 15–20 minutes in comparison to an equivalent IKD protocol while providing comparable reliability and validity. Practitioners should be sure to consistently test using the same weight stack machine when assessing patients over time, as different weight stack machines could produce slightly different results due to manufacture designs and specifications.

There were several limitations to this study. First, the major limitation of the ND compared to the IKD is that its application is limited to isometric testing. Second, our study is limited to the reliability and validity of knee extensor isometric strength. At the same time, we believe our findings will translate to testing other large muscles (e.g., hip extensors, hamstrings, hip abductors). However, our device was not tested on other muscle group specific weight stack machines. Third, given our participant directions to gradually build force during the test, we could not demonstrate our device's ability to provide accurate rate force development data. However, given the high correlation (and tight SD) between the ND force-time curves with that of the IKD, we see no reason why force-time curves from the ND would demonstrate a vastly different rate of force development value from those of the IKD. Fourth, our testers were all fit males. Thus, the findings of the HHD reliability and concurrent validity may not translate to all clinicians or fitness professionals using the HHD as described in our study.

Conclusion

The study supports both within- and between-day *interrater* test-retest reliability, and maximum force and force-time curve validity of this ND and testing paradigm. The findings suggest that when testing knee extensor force production, the ND is comparable to an IKD and may be superior to HHD in healthy individuals and those diagnosed with mild to moderate knee OA. The HHD protocol used in this study resulted in a ceiling effect, unlike the ND and IKD, indicating that using an HHD to measure larger muscle groups may not fully represent a participant's strength and is dependent on the resistance offered by the tester. Excellent device test-retest reliability and validity, portability, safety, and the hands-free protocol made possible by the ND suggest potential utility within physical therapy and fitness settings for strength testing large muscle groups such as knee extensors.

Acknowledgments

We want to acknowledge Trey Brindle, PhD, and Akil Piggott, DPT, who were integral in participant recruitment and data collection, and Sri Balasubramanian, PhD, a collaborator and fundamental part of the device development team.

Author Contributions

Conceptualization: Noel Goodstadt, Arun Ramakrishnan, Sheri P. Silfies.

Data curation: Pradeep K. Sahu, Sheri P. Silfies.

Formal analysis: Pradeep K. Sahu, Arun Ramakrishnan, Sheri P. Silfies.

Funding acquisition: Noel Goodstadt, Arun Ramakrishnan, Sheri P. Silfies.

Investigation: Noel Goodstadt.

Methodology: Noel Goodstadt, Arun Ramakrishnan, Sheri P. Silfies.

Software: Arun Ramakrishnan.

Validation: Pradeep K. Sahu, Noel Goodstadt, Arun Ramakrishnan, Sheri P. Silfies.

Visualization: Pradeep K. Sahu, Arun Ramakrishnan, Sheri P. Silfies.

Writing – original draft: Pradeep K. Sahu, Sheri P. Silfies.

Writing – review & editing: Pradeep K. Sahu, Noel Goodstadt, Arun Ramakrishnan, Sheri P. Silfies.

References

1. Majewski M., Susanne H., and Klaus S., Epidemiology of athletic knee injuries: A 10-year study. *Knee*, 2006. 13(3): p. 184–8. <https://doi.org/10.1016/j.knee.2006.01.005> PMID: 16603363
2. Joseph A.M., et al., A multisport epidemiologic comparison of anterior cruciate ligament injuries in high school athletics. *J Athl Train*, 2013. 48(6): p. 810–7. <https://doi.org/10.4085/1062-6050-48.6.03> PMID: 24143905
3. Ward S.H., et al., Quadriceps Neuromuscular Function and Jump-Landing Sagittal-Plane Knee Biomechanics After Anterior Cruciate Ligament Reconstruction. *J Athl Train*, 2018. 53(2): p. 135–143. <https://doi.org/10.4085/1062-6050-306-16> PMID: 29350554
4. Segal N.A., et al., Effect of quadriceps strength and proprioception on risk for knee osteoarthritis. *Med Sci Sports Exerc*, 2010. 42(11): p. 2081–8. <https://doi.org/10.1249/MSS.0b013e3181dd902e> PMID: 20351594
5. Larsen J.B., et al., Muscle strength and functional performance is markedly impaired at the recommended time point for sport return after anterior cruciate ligament reconstruction in recreational athletes. *Hum Mov Sci*, 2015. 39: p. 73–87. <https://doi.org/10.1016/j.humov.2014.10.008> PMID: 25461435
6. Karanikas K., Arampatzis A., and Brüggemann G.P., Motor task and muscle strength followed different adaptation patterns after anterior cruciate ligament reconstruction. *Eur J Phys Rehabil Med*, 2009. 45(1): p. 37–45. PMID: 18987563
7. Kamiya K., et al., Quadriceps Strength as a Predictor of Mortality in Coronary Artery Disease. *Am J Med*, 2015. 128(11): p. 1212–9. <https://doi.org/10.1016/j.amjmed.2015.06.035> PMID: 26169888
8. Swallow E.B., et al., Quadriceps strength predicts mortality in patients with moderate to severe chronic obstructive pulmonary disease. *Thorax*, 2007. 62(2): p. 115–20. <https://doi.org/10.1136/thx.2006.062026> PMID: 17090575
9. Chandler J.M., et al., Is lower extremity strength gain associated with improvement in physical performance and disability in frail, community-dwelling elders?. *Arch Phys Med Rehabil* 1998. 79: p. 24–30. [https://doi.org/10.1016/s0003-9993\(98\)90202-7](https://doi.org/10.1016/s0003-9993(98)90202-7) PMID: 9440412
10. Ringsberg K., et al., Is there a relationship between balance, gait performance and muscular strength in 75-year-old women? *Age and Ageing*, 1999. 28: p. 289–293. <https://doi.org/10.1093/ageing/28.3.289> PMID: 10475866

11. Davis J.W., et al., Strength, Physical Activity, and Body Mass Index: Relationship to Performance-Based Measures and Activities of Daily Li&g Among Older Japanese Women in Hawaii. *Journal of American Geriatrics Society*, 1998. 46: p. 274–279.
12. Wisloff U., Helgerud J., and Hoff J., Strength and endurance of elite soccer players. *Medicine & Science in Sports & Exercise*, 1998. 30(3): p. 462–467. <https://doi.org/10.1097/00005768-199803000-00019> PMID: 9526895
13. Kraemer W.J., et al., Effect of resistance training on women's strength/power and occupational performances. *Med. Sci. Sports Exerc*, 2001. 33: p. 1011–1025. <https://doi.org/10.1097/00005768-200106000-00022> PMID: 11404668
14. Beaudart C., et al., Assessment of Muscle Function and Physical Performance in Daily Clinical Practice: A position paper endorsed by the European Society for Clinical and Economic Aspects of Osteoporosis, Osteoarthritis and Musculoskeletal Diseases (ESCEO). *Calcif Tissue Int*, 2019. 105(1): p. 1–14. <https://doi.org/10.1007/s00223-019-00545-w> PMID: 30972475
15. Reilly T., Bangsbo J., and Franks A., Anthropometric and physiological predispositions for elite soccer. *J. Sports Sci*, 2000. 18: p. 669–683. <https://doi.org/10.1080/02640410050120050> PMID: 11043893
16. Burns S.P. and Spanier D.E., Break-technique handheld dynamometry: relation between angular velocity and strength measurements. *Arch Phys Med Rehabil*, 2005. 86(7): p. 1420–6. <https://doi.org/10.1016/j.apmr.2004.12.041> PMID: 16003675
17. Harbo T., Brincks J., and Andersen H., Maximal isokinetic and isometric muscle strength of major muscle groups related to age, body mass, height, and sex in 178 healthy subjects. *Eur J Appl Physiol*, 2012. 112(1): p. 267–75. <https://doi.org/10.1007/s00421-011-1975-3> PMID: 21537927
18. Weininger S., Kapur K.C., and Pecht M., Exploring Medical Device Reliability and Its Relationship to Safety and Effectiveness. *IEEE Transactions on Components and Packaging Technologies*, 2010. 33(1): p. 240–245.
19. Bannigan K. and Watson R., Reliability and validity in a nutshell. *J Clin Nurs*, 2009. 18(23): p. 3237–43. <https://doi.org/10.1111/j.1365-2702.2009.02939.x> PMID: 19930083
20. Bohannon R.W., Hand-held dynamometry: Adoption 1900–2005. *Percept Mot Skills*, 2006. 103: p. 3–4. <https://doi.org/10.2466/pms.103.1.3-4> PMID: 17037637
21. Wikholm J.B. and Bohannon R.W., Hand-held Dynamometer Measurements: Tester Strength Makes a Difference. *The Journal of orthopaedic and sports physical therapy*, 1991. 13(4): p. 191–198. <https://doi.org/10.2519/jospt.1991.13.4.191> PMID: 18796845
22. Brinkmann J.R., Comparison of a hand-held and fixed dynamometer in measuring strength of patients with neuromuscular disease. *Orthop Sports Phys Ther* 1994. 19: p. 100–104. <https://doi.org/10.2519/jospt.1994.19.2.100> PMID: 8148862
23. Bohannon R.W., Reference Values for Extremity Muscle Strength Obtained by Hand-Held Dynamometry From Adults Aged 20 to 79 Years. *Arch Phys Med Rehabil*, 1997. 78(9): p. 26–32. [https://doi.org/10.1016/s0003-9993\(97\)90005-8](https://doi.org/10.1016/s0003-9993(97)90005-8) PMID: 9014953
24. Whiteley R., et al., Correlation of isokinetic and novel hand-held dynamometry measures of knee flexion and extension strength testing. *J Sci Med Sport*, 2012. 15(5): p. 444–50. <https://doi.org/10.1016/j.jsams.2012.01.003> PMID: 22424705
25. Chamorro C., et al., Absolute Reliability and Concurrent Validity of Hand Held Dynamometry and Isokinetic Dynamometry in the Hip, Knee and Ankle Joint: Systematic Review and Meta-analysis. *Open Med (Wars)*, 2017. 12: p. 359–375. <https://doi.org/10.1515/med-2017-0052> PMID: 29071305
26. Lu T.W., et al., Enhancing the examiner's resisting force improves the reliability of manual muscle strength measurements: comparison of a new device with hand-held dynamometry. *J Rehabil Med*, 2007. 39(9): p. 679–84. <https://doi.org/10.2340/16501977-0107> PMID: 17999004
27. Kelln B.M., et al., Hand-Held Dynamometry: Reliability of Lower Extremity Muscle Testing in Healthy, Physically Active, Young Adults. *Journal of Sport Rehabilitation*, 2008. 17: p. 160–170.
28. de Araujo Ribeiro Alvares J.B., et al., Inter-machine reliability of the Biodex and Cybex isokinetic dynamometers for knee flexor/extensor isometric, concentric and eccentric tests. *Phys Ther Sport*, 2015. 16(1): p. 59–65. <https://doi.org/10.1016/j.ptsp.2014.04.004> PMID: 24913915
29. Toonstra J. and Mattacola C.G., Test-retest reliability and validity of isometric knee-flexion and -extension measurement using 3 methods of assessing muscle strength. *J Sport Rehabil*, 2013. 22(1). <https://doi.org/10.1123/jsr.2013.TR7> PMID: 22951307
30. Mendoza M. and Miller R.G., Muscle Strength, Assessment of, in *Encyclopedia of the Neurological Sciences* (Second Edition), Aminoff M.J. and Daroff R.B., Editors. 2014, Academic Press: Oxford. p. 190–193.
31. Mavroidis C., et al., Smart portable rehabilitation devices. *J Neuroeng Rehabil*, 2005. 2: p. 18. <https://doi.org/10.1186/1743-0003-2-18> PMID: 16011801

32. Mentiplay B.F., et al., Assessment of Lower Limb Muscle Strength and Power Using Hand-Held and Fixed Dynamometry: A Reliability and Validity Study. *PLoS One*, 2015. 10(10): p. e0140822. <https://doi.org/10.1371/journal.pone.0140822> PMID: 26509265
33. Morin M., et al., Psychometric properties of a standardized protocol of muscle strength assessment by hand-held dynamometry in healthy adults: a reliability study. *BMC Musculoskelet Disord*, 2023. 24(1): p. 294. <https://doi.org/10.1186/s12891-023-06400-2> PMID: 37060020
34. Hansen E.M., et al., Hand-held Dynamometer Positioning Impacts Discomfort During Quadriceps Strength Testing: A Validity and Reliability Study. *Int J Sports Phys Ther*, 2015. 10(1): p. 62–68. PMID: 25709864
35. Gagnon D., et al., Reliability and validity of static knee strength measurements obtained with a chair-fixed dynamometer in subjects with hip or knee arthroplasty. *Arch Phys Med Rehabil*, 2005. 86(10): p. 1998–2008. <https://doi.org/10.1016/j.apmr.2005.04.013> PMID: 16213245
36. Martins J., et al., Reliability and Validity of the Belt-Stabilized Handheld Dynamometer in Hip- and Knee-Strength Tests. *J Athl Train*, 2017. 52(9): p. 809–819. <https://doi.org/10.4085/1062-6050-52.6.04> PMID: 28787180
37. Lesnak J., et al., Validity of Hand-Held Dynamometry in Measuring Quadriceps Strength and Rate of Torque Development. *International Journal of Sports Physical Therapy*, 2019. 14(2): p. 180–187. PMID: 30997270
38. Lipovšek T., Kacin A., and Puh U., Reliability and validity of hand-held dynamometry for assessing lower limb muscle strength. *Isokinetics and Exercise Science*, 2022. 30(3): p. 231–240.
39. Jackson S.M., et al., Intrarater reliability of hand held dynamometry in measuring lower extremity isometric strength using a portable stabilization device. *Musculoskelet Sci Pract*, 2017. 27: p. 137–141. <https://doi.org/10.1016/j.math.2016.07.010> PMID: 27476066
40. Koblbauer I.F., et al., Reliability of maximal isometric knee strength testing with modified hand-held dynamometry in patients awaiting total knee arthroplasty: useful in research and individual patient settings? A reliability study. *BMC Musculoskelet Disord*, 2011. 12: p. 249. <https://doi.org/10.1186/1471-2474-12-249> PMID: 22040119
41. Sung K.S., Yi Y.G., and Shin H.I., Reliability and validity of knee extensor strength measurements using a portable dynamometer anchoring system in a supine position. *BMC Musculoskelet Disord*, 2019. 20(1): p. 320. <https://doi.org/10.1186/s12891-019-2703-0> PMID: 31286912
42. Martin H.J., et al., Is hand-held dynamometry useful for the measurement of quadriceps strength in older people? A comparison with the gold standard Bodex dynamometry. *Gerontology*, 2006. 52(3): p. 154–9. <https://doi.org/10.1159/000091824> PMID: 16645295
43. Bohannon R.W., et al., Adequacy of belt-stabilized testing of knee extension strength. *Journal of strength and conditioning research*, 2011. 25(7): p. 1963–1967. <https://doi.org/10.1519/JSC.0b013e3181e4f5ce> PMID: 21399535
44. Katoh M. and Yamasaki H., Test-retest reliability of isometric leg muscle strength measurements made using a hand-held dynamometer restrained by a belt: comparisons during and between sessions. *J Phys Ther Sci*, 2009(21): p. 239–43.
45. Douma K.W., et al., Reliability of the Q Force; a mobile instrument for measuring isometric quadriceps muscle strength. *BMC Sports Sci Med Rehabil*, 2016. 8: p. 4. <https://doi.org/10.1186/s13102-016-0029-x> PMID: 26900478
46. Verkerke G.J., et al., Precision, comfort and mechanical performance of the Quadriso-tester, a quadriceps force measuring device. 2003. 41(3): p. 283–289. <https://doi.org/10.1007/BF02348432> PMID: 12803292
47. Hartog J., et al., A portable isometric knee extensor strength testing device: test-retest reliability and minimal detectable change scores of the Q-Force II in healthy adults. *BMC Musculoskelet Disord*, 2021. 22(1): p. 966.
48. O'Brien M., et al., A novel device to assess hip strength: Concurrent validity and normative values in male athletes. *Phys Ther Sport*, 2019. 35: p. 63–68. <https://doi.org/10.1016/j.ptsp.2018.11.006> PMID: 30471548
49. Bazett-Jones D.M. and Squier K., Measurement properties of hip strength measured by handheld dynamometry: Reliability and validity across the range of motion. *Phys Ther Sport*, 2020. 42: p. 100–106. <https://doi.org/10.1016/j.ptsp.2020.01.005> PMID: 31958744
50. Dunne C., et al., Validity, reliability, minimal detectable change, and methodological considerations for HHD and portable fixed frame isometric hip and groin strength testing: A comparison of unilateral and bilateral testing methods. *Phys Ther Sport*, 2022. 57: p. 46–52. <https://doi.org/10.1016/j.ptsp.2022.07.002> PMID: 35921781

51. Stark T., et al., Hand-held dynamometry correlation with the gold standard isokinetic dynamometry: a systematic review. *PM R*, 2011. 3(5): p. 472–9. <https://doi.org/10.1016/j.pmrj.2010.10.025> PMID: 21570036
52. Schrama P.P., et al., Intraexaminer reliability of hand-held dynamometry in the upper extremity: a systematic review. *Arch Phys Med Rehabil*, 2014. 95(12): p. 2444–69. <https://doi.org/10.1016/j.apmr.2014.05.019> PMID: 24909587
53. Walter S.D., Eliasziw M., and Donner A., Sample size and optimal designs for reliability studies. *Statistics in Medicine*, 1998. 17(1): p. 101–110. [https://doi.org/10.1002/\(sici\)1097-0258\(19980115\)17:1<101::aid-sim727>3.0.co;2-e](https://doi.org/10.1002/(sici)1097-0258(19980115)17:1<101::aid-sim727>3.0.co;2-e) PMID: 9463853
54. Borg D.N., et al., Calculating sample size for reliability studies. *PM R*, 2022. 14(8): p. 1018–1025. <https://doi.org/10.1002/pmrj.12850> PMID: 35596122
55. Saltin B. and Grimby G., Physiological analysis of middle-aged and old former athletes Comparison with still active athletes of the same ages. *Circulation*, 1968. 38(6): p. 1104–1115.
56. Lindwall M., et al., The relationships of change in physical activity with change in depression, anxiety, and burnout: a longitudinal study of Swedish healthcare workers. *Health Psychol*, 2014. 33(11): p. 1309–18. <https://doi.org/10.1037/a0034402> PMID: 24245832
57. Rodger L., et al., Self-reported leisure time physical activity: a useful assessment tool in everyday health care. *BMC Public Health*, 2012(12): p. 693. <https://doi.org/10.1186/1471-2458-12-693> PMID: 22920914
58. Apullan F.J., et al., Usefulness of self-reported leisure-time physical activity to predict long-term survival in patients with coronary heart disease. *Am J Cardiol*, 2008. 102(4): p. 375–9. <https://doi.org/10.1016/j.amjcard.2008.03.072> PMID: 18678290
59. Aires N., Selmer R., and Thelle D., The validity of self-reported leisure time physical activity, and its relationship to serum cholesterol, blood pressure and body mass index. A population based study of 332,182 men and women aged 40–42 years. *Eur J Epidemiol*, 2003. 18(6): p. 479–485. <https://doi.org/10.1023/a:1024682523710> PMID: 12908712
60. Roach K.E., et al., Test-retest reliability of patient reports of low back pain. *The Journal of Orthopaedic and Sports Physical Therapy*, 1997. 26: p. 253–259. <https://doi.org/10.2519/jospt.1997.26.5.253> PMID: 9353688
61. Moudgil R. and Karpovich P.V., Duration of a Maximal Isometric Muscular Contraction. *Research Quarterly. American Association for Health, Physical Education and Recreation*, 2013. 40(3): p. 536–539.
62. Astrand P.O. and Rodahl K., A text book of work physiology: Physiological bases of exercise. 2nd Edition ed. 1977, New York: McGraw-Hill Book Company.
63. Weir J.P., Quantifying test-retest reliability using the intraclass correlation coefficient and the SEM. *Journal of strength and conditioning research*, 2005. 19(1): p. 231–240. <https://doi.org/10.1519/15184.1> PMID: 15705040
64. Koo T.K. and Li M.Y., A Guideline of Selecting and Reporting Intraclass Correlation Coefficients for Reliability Research. *J Chiropr Med*, 2016. 15(2): p. 155–63. <https://doi.org/10.1016/j.jcm.2016.02.012> PMID: 27330520
65. Bland J.M. and Altman D.G., Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet*, 1986. 1(8476): p. 307–10. PMID: 2868172
66. Schober P., Boer C., and Schwarte L.A., Correlation Coefficients: Appropriate Use and Interpretation. *Anesth Analg*, 2018. 126(5): p. 1763–1768. <https://doi.org/10.1213/ANE.0000000000002864> PMID: 29481436
67. McHugh M.L., Interrater reliability: the kappa statistic. *Biochem Med (Zagreb)*, 2012. 22(3): p. 276–282. <https://doi.org/10.1016/j.jocd.2012.03.005> PMID: 23092060
68. Tuijn S., et al., Reducing interrater variability and improving health care: a meta-analytical review. *J Eval Clin Pract*, 2012. 18(4): p. 887–95. <https://doi.org/10.1111/j.1365-2753.2011.01705.x> PMID: 21726359
69. Arnold C.M., et al., The reliability and validity of handheld dynamometry for the measurement of lower-extremity muscle strength in older adults. *J Strength Cond Res*, 2010. 24(3): p. 815–824. <https://doi.org/10.1519/JSC.0b013e3181aa36b8> PMID: 19661831
70. Knols R.H., et al., Hand-held dynamometry in patients with haematological malignancies: measurement error in the clinical assessment of knee extension strength. *BMC Musculoskelet Disord*, 2009. 10: p. 31. <https://doi.org/10.1186/1471-2474-10-31> PMID: 19272149
71. Pinto-Ramos J., et al., Handheld dynamometer reliability to measure knee extension strength in rehabilitation patients-A cross-sectional study. *PLoS One*, 2022. 17(5): p. e0268254. <https://doi.org/10.1371/journal.pone.0268254> PMID: 35580110

72. Angelozzi M., et al., Rate of force development as an adjunctive outcome measure for return-to-sport decisions after anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*, 2012. 42(9): p. 772–80. <https://doi.org/10.2519/jospt.2012.3780> PMID: [22814219](https://pubmed.ncbi.nlm.nih.gov/22814219/)